DemoBank QA audit

Target **http://localhost:5173** · API http://localhost:8000 · 2026-10-21T16:38:46 · 197.9 s · LLM groq (openai/gpt-oss-120b / openai/gpt-oss-120b)

31	5	0	42
requirements passed	requirements failed	unverified	scenarios · 190 assertions

$0.0005

Claude-equivalent cost · Groq list price $0.0007 (free tier: $0 billed)

Executive summary

Five scenarios failed due to data mismatches and validation gaps. History rows show zero change instead of actual amounts, and date-boundary filtering omits expected transactions. Transfer endpoint incorrectly accepts overdraft, negative, and zero amounts, creating records and altering balances. These indicate missing validation and incorrect display logic.

Defects (6)

D1 · Critical **Reject a negative amount ('-5.00')** negative TRF-AMT-NEG

Requirements: TRF-04, TRF-08

▼ **Steps to reproduce**

1. Open the app in a fresh browser profile and click 'Enter Demo'
2. Transfer form: from 'Everyday Checking' to 'Rainy Day Savings', amount '-5.00', memo 'QA-163846-TRF-AMT-NEG-amt1' → submit

a negative amount ('-5.00'): the transfer is rejected [TRF-04]

Expected	Observed
validation/error message, no success	

	SUCCESS shown; API 201; messages: Transfer complete Amount −$5.00 From Everyday Checking (•••• 1042) To Rainy Day Savings (•••• 7781) Date & time Oct 21, 2026, 1:40 PM PDT Memo QA-163846-TRF-AMT

a negative amount ('-5.00'): no success state is shown [TRF-08]

Expected	Observed
no 'Transfer complete'	SUCCESS shown; API 201; messages: Transfer complete Amount −$5.00 From Everyday Checking (•••• 1042) To Rainy Day Savings (•••• 7781) Date & time Oct 21, 2026, 1:40 PM PDT Memo QA-163846-TRF-AMT

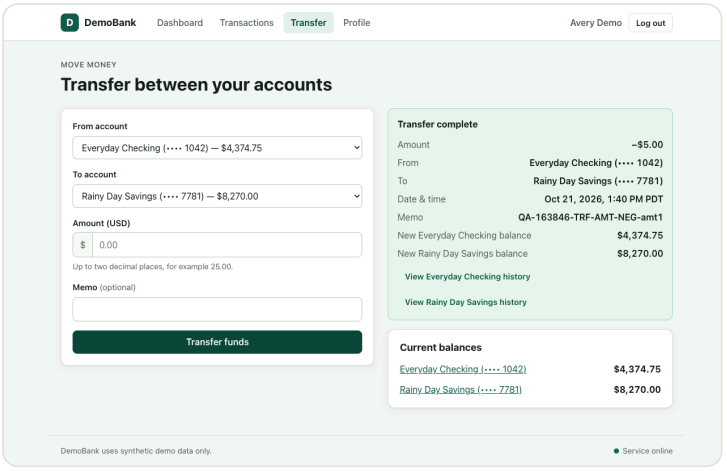
A negative amount ('-5.00'): balances are unchanged [TRF-04]

Expected	Observed
both balances unchanged	{'Everyday Checking': '$4369.75 → $4374.75', 'Rainy Day Savings': '$8275.00 → $8270.00'}

A negative amount ('-5.00'): no transaction record is created [TRF-04]

Expected	Observed
0 new records	transfer completed $-5.00

> **Likely cause / fix:** Negative amount not validated; reject amounts <=0 with error response.



▶ **Observed values & network evidence**

D2 · Critical **Reject zero ('0')** negative TRF-AMT-ZERO

Requirements: TRF-04, TRF-08

▼ **Steps to reproduce**

1. Open the app in a fresh browser profile and click 'Enter Demo'

2. Transfer form: from 'Everyday Checking' to 'Rainy Day Savings', amount '0', memo 'QA-163846-TRF-AMT-ZERO-amt1' → submit

zero ('0'): the transfer is rejected [TRF-04]

Expected	Observed
validation/error message, no success	SUCCESS shown; API 201; messages: Transfer complete Amount $0.00 From Everyday Checking (•••• 1042) To Rainy Day Savings (•••• 7781) Date & time Oct 21, 2026, 1:40 PM PDT Memo QA-163846-TRF-AMT-

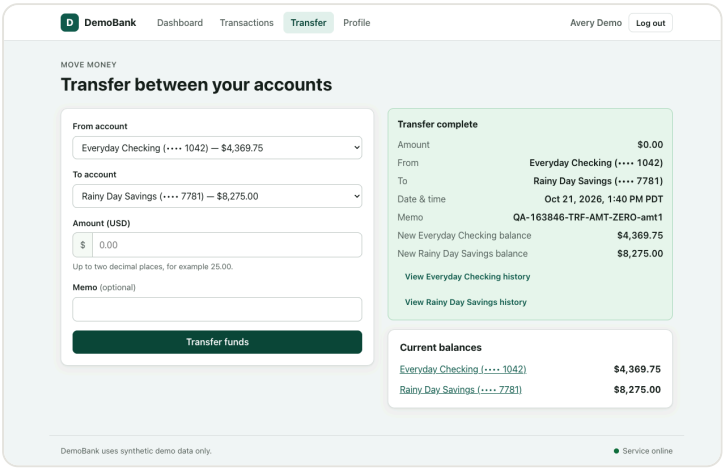
zero ('0'): no success state is shown [TRF-08]

Expected	Observed
no 'Transfer complete'	SUCCESS shown; API 201; messages: Transfer complete Amount $0.00 From Everyday Checking (•••• 1042) To Rainy Day Savings (•••• 7781) Date & time Oct 21, 2026, 1:40 PM PDT Memo QA-163846-TRF-AMT-

Zero ('0'): no transaction record is created [TRF-04]

Expected	Observed
0 new records	transfer completed $0.00

> **Likely cause / fix:** Zero amount not validated; reject and do not create transaction record.



▶ **Observed values & network evidence**

D3 · Critical **Reject zero ('0.00')** edge TRF-AMT-ZERO2

Requirements: TRF-04, TRF-08

▼ **Steps to reproduce**

1. Open the app in a fresh browser profile and click 'Enter Demo'

2. Transfer form: from 'Everyday Checking' to 'Rainy Day Savings', amount '0.00', memo 'QA-163846-TRF-AMT-ZERO2-amt1' → submit

zero ('0.00'): the transfer is rejected [TRF-04]

Expected	Observed
validation/error message, no success	SUCCESS shown; API 201; messages: Transfer complete Amount $0.00 From Everyday Checking (•••• 1042) To Rainy Day Savings (•••• 7781) Date & time Oct 21, 2026, 1:40 PM PDT Memo QA-163846-TRF-AMT-

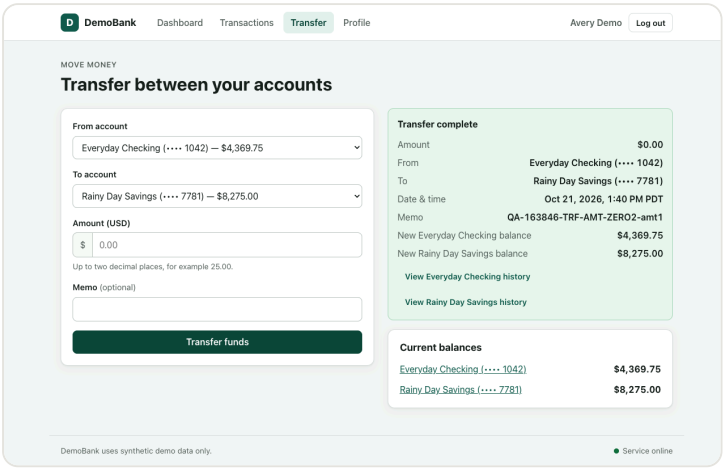
zero ('0.00'): no success state is shown [TRF-08]

Expected	Observed
no 'Transfer complete'	SUCCESS shown; API 201; messages: Transfer complete Amount $0.00 From Everyday Checking (•••• 1042) To Rainy Day Savings (•••• 7781) Date & time Oct 21, 2026, 1:40 PM PDT Memo QA-163846-TRF-AMT-

Zero ('0.00'): no transaction record is created [TRF-04]

Expected	Observed
0 new records	transfer completed $0.00

> **Likely cause / fix:** Same as TRF-AMT-ZERO – ensure zero-value transfers are blocked.



▶ **Observed values & network evidence**

D4 · Critical **Amount one cent above the available balance is rejected** negative TRF-OVERDRAFT

Requirements: TRF-05, TRF-08

▼ **Steps to reproduce**

1. Open the app in a fresh browser profile and click 'Enter Demo'

2. Transfer form: from 'Everyday Checking' to 'Rainy Day Savings', amount '4369.76', memo 'QA-163846-TRF-OVERDRAFT-over1' → submit

amount $4369.76 (balance + $0.01): the transfer is rejected [TRF-05]

Expected	Observed
validation/error message, no success	SUCCESS shown; API 201; messages: Transfer complete Amount $4,369.76 From Everyday Checking (•••• 1042) To Rainy Day Savings (•••• 7781) Date & time Oct 21, 2026, 1:40 PM PDT Memo QA-163846-TRF-

amount $4369.76 (balance + $0.01): no success state is shown [TRF-08]

Expected	Observed
no 'Transfer complete'	SUCCESS shown; API 201; messages: Transfer complete Amount $4,369.76 From Everyday Checking (•••• 1042) To Rainy Day Savings (•••• 7781) Date & time Oct 21, 2026, 1:40 PM PDT Memo QA-163846-TRF-

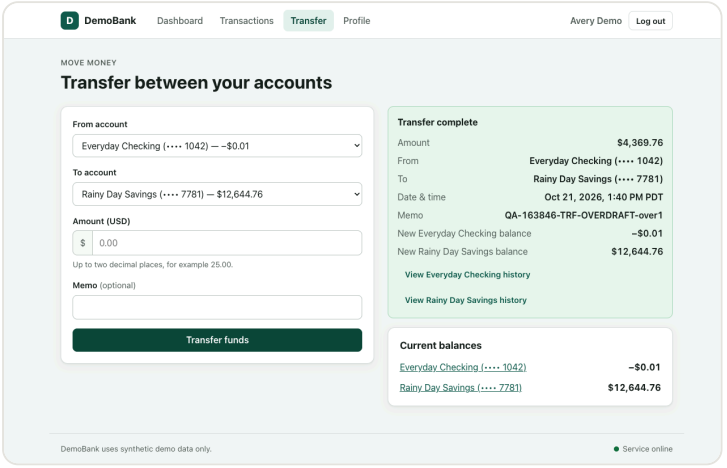
Insufficient funds: balances are unchanged [TRF-05]

Expected	Observed
both balances unchanged	{'Everyday Checking': '$4369.75 → $-0.01', 'Rainy Day Savings': '$8275.00 → $12644.76'}

Insufficient funds: no transaction record is created [TRF-05]

Expected	Observed
0 new records	transfer completed $4369.76

> **Likely cause / fix:** Transfer service lacks overdraft check; enforce balance >= amount before processing.



▶ **Observed values & network evidence**

D5 · Medium **Start and end dates are inclusive (including Pacific-time edge cases)** edge HIS-DATE-BOUNDARY

Requirements: HIS-04

▼ **Steps to reproduce**

1. Open the app in a fresh browser profile and click 'Enter Demo'
2. Filter dates: start = end = that day [2025-03-04 → 2025-03-04] and look for Demo utilities (2025-03-05T02:45:00Z UTC = 2025-03-04 PT)
3. Filter dates: start = that day [2025-03-04 → ...] and look for Demo utilities (2025-03-05T02:45:00Z UTC = 2025-03-04 PT)
4. Filter dates: end = that day [... → 2025-03-04] and look for Demo utilities (2025-03-05T02:45:00Z UTC = 2025-03-04 PT)
5. Filter dates: start = end = that day [2026-10-21 → 2026-10-21] and look for QA-162740 restore after TRF-MEMO-TEXT (2026-10-21T20:30:10.540817Z UTC = 2026-10-21 PT)
6. Filter dates: start = that day [2026-10-21 → ...] and look for QA-162740 restore after TRF-MEMO-TEXT (2026-10-21T20:30:10.540817Z UTC = 2026-10-21 PT)
7. Filter dates: end = that day [... → 2026-10-21] and look for QA-162740 restore after TRF-MEMO-TEXT (2026-10-21T20:30:10.540817Z UTC = 2026-10-21 PT)
8. Filter dates: start = end = that day [2026-10-21 → 2026-10-21] and look for bold & "quote" QA-162740-TRF-MEMO-TEXT-html1 (2026-10-21T20:30:08.853899Z UTC = 2026-10-21 PT)
9. Filter dates: start = that day [2026-10-21 → ...] and look for bold & "quote" QA-162740-TRF-MEMO-TEXT-html1 (2026-10-21T20:30:08.853899Z UTC = 2026-10-21 PT)
10. Filter dates: end = that day [... → 2026-10-21] and look for bold & "quote" QA-162740-TRF-MEMO-TEXT-html1 (2026-10-21T20:30:08.853899Z UTC = 2026-10-21 PT)

start = end = that day: includes Demo utilities (2025-03-05T02:45:00Z UTC = 2025-03-04 PT) [HIS-04]

Expected	Observed
transaction included	missing (0 rows shown)

end = that day: includes Demo utilities (2025-03-05T02:45:00Z UTC = 2025-03-04 PT) [HIS-04]

Expected	Observed
transaction included	missing (4 rows shown)

start = end = that day: includes QA-162740 restore after TRF-MEMO-TEXT (2026-10-21T20:30:10.540817Z UTC = 2026-10-21 PT) [HIS-04]

Expected	Observed
transaction included	missing (0 rows shown)

end = that day: includes QA-162740 restore after TRF-MEMO-TEXT (2026-10-21T20:30:10.540817Z UTC = 2026-10-21 PT) [HIS-04]

Expected	Observed
transaction included	missing (12 rows shown)

start = end = that day: includes bold & "quote" QA-162740-TRF-MEMO-TEXT-html1 (2026-10-21T20:30:08.853899Z UTC = 2026-10-21 PT) [HIS-04]

Expected	Observed
transaction included	missing (0 rows shown)

end = that day: includes bold & "quote" QA-162740-TRF-MEMO-TEXT-html1 (2026-10-21T20:30:08.853899Z UTC = 2026-10-21 PT) [HIS-04]

Expected	Observed
transaction included	missing (12 rows shown)

> **Likely cause / fix:** Date filter excludes edge-case timestamps; treat start/end as inclusive with proper timezone conversion.

D6 · Medium **Combined history shows every transaction with complete, correct row details** positive HIS-ROWS

Requirements: HIS-02

▼ **Steps to reproduce**

1. Open the app in a fresh browser profile and click 'Enter Demo'
2. Open /transactions without filters and read every table row

Each row's amount equals the transaction amount [HIS-02]

Expected	Observed
row amount = record amount	["Returned demo deposit (failed deposit): shows 'No change:\\n$0.00', record amount $700.00", "Pending demo debit (pending withdrawal): shows 'No change:\\n$0.00', record amount $125.00", "Declined demo item (failed withdrawal): shows 'No change:\\n$0.00', record amount $900.00", "Pending demo deposit (pending deposit): shows 'No change:\\n$0.00', record amount $400.00"]

> **Likely cause / fix:** Row rendering uses placeholder 'No change' instead of actual amount; map record amount to UI field.

Requirement coverage

ID	Requirement	Verdict	Why
GEN-01	The application must never initiate real payments or connect to an external bank.	PASS	1 assertion(s) passed across 1 scenario(s)
SES-01	The target has one seeded demo customer.	PASS	1 assertion(s) passed across 1 scenario(s)
SES-02	The landing page provides a clearly labeled Enter Demo action. It opens the demo dashboard without requiring a password or real identity information.	PASS	3 assertion(s) passed across 1 scenario(s)
SES-03	The header identifies the active demo customer and provides a Log out action.	PASS	2 assertion(s) passed across 1 scenario(s)
SES-04	Logging out returns the user to the landing page. Protected dashboard views should not remain usable after logout; direct navigation to them should return the user to the landing page or an access-required state.	PASS	12 assertion(s) passed across 2 scenario(s)
SES-05	Refreshing while in a valid demo session should preserve that session. A logged-out user must not regain access merely by refreshing a protected page.	PASS	4 assertion(s) passed across 2 scenario(s)
DSH-01	The dashboard shows the customer's checking and savings accounts.	PASS	1 assertion(s) passed across 1 scenario(s)
DSH-02	Each account displays a readable account name, a masked account identifier, account type, and current available balance in USD.	PASS	2 assertion(s) passed across 1 scenario(s)
DSH-03	Balances are displayed to two decimal places. The UI must not expose a full account identifier.	PASS	8 assertion(s) passed across 2 scenario(s)
DSH-04	Selecting an account opens an account detail view with its current balance and related transaction history.	PASS	2 assertion(s) passed across 1 scenario(s)

DSH-05	Account balances shown on the dashboard and detail view must agree with the latest successful transactions.	PASS	5 assertion(s) passed across 3 scenario(s)
TRF-01	The user can transfer funds between the checking and savings accounts owned by the demo customer.	PASS	2 assertion(s) passed across 2 scenario(s)
TRF-02	The transfer form includes source account, destination account, amount, and an optional memo.	PASS	3 assertion(s) passed across 2 scenario(s)
TRF-03	The source and destination must be different accounts. Same-account transfers must be rejected with a clear explanation and must not alter balances or create a transaction.	PASS	4 assertion(s) passed across 1 scenario(s)
TRF-04	Amount must be a positive number greater than zero, with no more than two digits after the decimal point. Blank, zero, negative, non-numeric, and over-precision values must be rejected with a clear validation message. Rejected input must not create a transaction or change balances.	FAIL	7 failing assertion(s), e.g. a negative amount ('-5.00'): the transfer is rejected
TRF-05	The amount must not exceed the source account's available balance. If funds are insufficient, the transfer must be rejected, both balances must remain unchanged, and no completed transfer must appear in history.	FAIL	3 failing assertion(s), e.g. amount $4369.76 (balance + $0.01): the transfer is rejected
TRF-06	A valid transfer is atomic: the source balance decreases by the amount and the destination balance increases by the same amount. No partial transfer is allowed.	PASS	3 assertion(s) passed across 3 scenario(s)
TRF-07	A successful transfer creates one completed transfer record that can be found in both relevant account histories. The record shows the amount, date and time, source, destination, status, and memo if supplied.	PASS	5 assertion(s) passed across 2 scenario(s)
TRF-08	The UI reports success only after the transfer has been committed. If a transfer is rejected or fails, the UI must explain that it did not complete and must not show a success state.	FAIL	4 failing assertion(s), e.g. amount $4369.76 (balance + $0.01): no success state is shown
TRF-09	Refreshing after a successful transfer must preserve the updated balances and transaction record.	PASS	1 assertion(s) passed across 1 scenario(s)

HIS-01	The user can view a combined transaction history and account-specific history.	PASS	5 assertion(s) passed across 2 scenario(s)
HIS-02	Each row shows date and time, description or memo, transaction type, amount, status, and relevant account. Amount direction should be clear for deposits, withdrawals, and transfers.	FAIL	1 failing assertion(s), e.g. Each row's amount equals the transaction amount
HIS-03	The user can filter history by account, transaction type, status, and date range. When filters are applied, every displayed row must satisfy all selected filters.	PASS	21 assertion(s) passed across 3 scenario(s)
HIS-04	The start and end dates are inclusive. A transaction occurring on either boundary date must be included when it matches the other filters.	FAIL	6 failing assertion(s), e.g. start = end = that day: includes Demo utilities (2025-03-05T02:45:00Z UTC = 2025-03-04 PT)
HIS-05	An empty result set must show a clear empty state, not an error or unrelated transactions.	PASS	2 assertion(s) passed across 1 scenario(s)
HIS-06	Clearing filters restores the unfiltered history.	PASS	2 assertion(s) passed across 1 scenario(s)
PRF-01	A profile view displays only synthetic demo-customer details, such as display name and a masked email address.	PASS	2 assertion(s) passed across 1 scenario(s)
PRF-02	The view must not reveal passwords, authentication tokens, full account identifiers, or data belonging to another user.	PASS	2 assertion(s) passed across 1 scenario(s)
UI-01	The target works in a current desktop browser at a 1280 × 800 viewport and remains usable at narrower widths.	PASS	9 assertion(s) passed across 1 scenario(s)
UI-02	Interactive controls have readable labels and communicate loading, success, validation, and failure states.	PASS	6 assertion(s) passed across 2 scenario(s)
UI-03	A submitted action should not silently disappear. While waiting, the UI should show a loading state or otherwise prevent accidental duplicate submission.	PASS	2 assertion(s) passed across 1 scenario(s)
UI-04	Navigation links and buttons lead to the labeled destination or perform the labeled action.	PASS	8 assertion(s) passed across 1 scenario(s)

UI-05	A recoverable error must leave the user able to retry or return to the dashboard.	PASS	5 assertion(s) passed across 3 scenario(s)
SEED-01	The demo begins with one customer, one checking account, one savings account, and enough historical transactions to exercise account, type, status, and date filters.	PASS	2 assertion(s) passed across 1 scenario(s)
SEED-02	Seed records should include multiple dates and at least one transfer between the two accounts.	PASS	1 assertion(s) passed across 1 scenario(s)
SEED-03	The demo has no signup, external payment, bill-pay, loan, card, or multi-user features.	PASS	1 assertion(s) passed across 1 scenario(s)

All scenarios

▶ PASS **Seed data: one customer, one checking + one savings, varied history incl. a transfer** positive SEED-DATA 1.8s

▶ PASS **Enter Demo opens the dashboard without credentials** positive SES-ENTER 2.7s

▶ PASS **Header shows the active customer and a Log out action** positive SES-HEADER 1.7s

▶ PASS **Every demo session is the same single seeded customer** positive SES-SINGLE 3.4s

▶ PASS **Refreshing protected pages keeps the session** positive SES-REFRESH 7.2s

▶ PASS **A visitor who never entered the demo cannot open protected pages** negative SES-ANON 4.0s

▶ PASS **Dashboard lists checking and savings with masked IDs and USD balances** positive DSH-CARDS 2.7s

▶ PASS **Selecting each account opens a detail view with balance and its own history** positive DSH-DETAIL 5.0s

▶ PASS **Masked identifier format is consistent across dashboard, detail and forms** edge DSH-MASK-API 4.4s

▶ FAIL **Combined history shows every transaction with complete, correct row details** positive HIS-ROWS 2.8s

▶ PASS **Each single filter (account, type, status) shows exactly the matching rows** positive HIS-FILTER-SINGLE 6.3s

- PASS Combined filters (account + type + status + dates) intersect correctly
  edge HIS-FILTER-COMBO 3.1s
- FAIL Start and end dates are inclusive (including Pacific-time edge cases)
  edge HIS-DATE-BOUNDARY 13.1s
- PASS A filter with no matches shows a clear empty state edge HIS-EMPTY 3.1s
- PASS Clearing filters restores the full history positive HIS-CLEAR 3.6s
- PASS Start date after end date gives a clear message (no misleading rows)
  negative HIS-DATE-INVERTED 3.1s
- PASS Profile shows display name and a masked email only positive PRF-VIEW 2.7s
- PASS No passwords, tokens or full account numbers leak into the page or storage
  negative PRF-SECRETS 2.6s
- PASS Transfer form has source, destination, amount and optional memo
  positive TRF-FORM 2.7s
- PASS Valid transfer with memo: balances, records, histories and refresh all agree
  positive TRF-VALID 10.1s
- PASS Valid transfer without memo (memo is optional), savings → checking
  positive TRF-VALID-NOMEMO 3.4s
- PASS Smallest valid amount ($0.01) is accepted and moved exactly edge TRF-MIN 3.4s
- PASS Transferring exactly the full available balance is allowed edge TRF-EXACT-BALANCE 3.4s
- FAIL Amount one cent above the available balance is rejected negative TRF-OVERDRAFT 3.4s
- PASS Same source and destination account is rejected negative TRF-SAME 3.5s
- PASS Reject a blank amount negative TRF-AMT-BLANK 3.4s
- PASS Reject a non-numeric amount ('abc') negative TRF-AMT-NAN 3.4s
- FAIL Reject a negative amount ('-5.00') negative TRF-AMT-NEG 3.4s
- PASS Reject three decimal places ('1.234') edge TRF-AMT-PREC 3.4s
- PASS Reject scientific notation ('1e2') edge TRF-AMT-SCI 3.4s
- PASS Reject whitespace only edge TRF-AMT-SPACE 3.4s
- FAIL Reject zero ('0') negative TRF-AMT-ZERO 3.4s
- FAIL Reject zero ('0.00') edge TRF-AMT-ZERO2 3.4s
- PASS A server failure is reported as not completed and the user can retry
  negative TRF-SERVER-FAIL 5.0s

▶ PASS **Double-clicking submit on a slow network creates only one transfer** edge TRF-DOUBLE 4.7s

▶ PASS **Memo with HTML/special characters is stored and shown literally** edge TRF-MEMO-TEXT 4.5s

▶ PASS **Usable at 1280×800 and at narrower widths (tablet, phone)** edge UI-RESPONSIVE 10.9s

▶ PASS **Every interactive control has a readable accessible label** positive UI-LABELS 5.3s

▶ PASS **Navigation links and buttons lead to their labelled destinations** positive UI-NAV 14.9s

▶ PASS **When account data fails to load, the user can retry or go back** negative UI-ERROR-RECOVERY 3.0s

▶ PASS **Log out returns to landing and locks protected routes** negative SES-LOGOUT 10.5s

▶ PASS **The app only talks to its own frontend and API hosts** positive GEN-NETWORK 1.7s

Cost & efficiency

Purpose	Model	Calls	Cached	Input tok	Output tok	USD (provider)	USD (Claude-equiv.)
prd_enrichment	openai/gpt-oss-120b	1	1	1,216	2,717	$0.0000	$0.0000
clarity_judge	openai/gpt-oss-120b	1	0	784	243	$0.0003	$0.0002
executive_summary	openai/gpt-oss-120b	1	0	1,512	289	$0.0004	$0.0003

Browser actions: 261 · API responses captured: 558 · Locator heals: 0 · Deterministic fallbacks: none

Test hygiene – compensating transfers made to restore balances:
compensating transfer 1.37 Rainy Day Savings → Everyday Checking: HTTP 201
compensating transfer 2.05 Everyday Checking → Rainy Day Savings: HTTP 201
compensating transfer 0.01 Rainy Day Savings → Everyday Checking: HTTP 201
compensating transfer 4369.75 Rainy Day Savings → Everyday Checking: HTTP 201
compensating transfer 4369.76 Rainy Day Savings → Everyday Checking: HTTP 201
compensating transfer 5.00 Everyday Checking → Rainy Day Savings: HTTP 201
compensating transfer 3.00 Rainy Day Savings → Everyday Checking: HTTP 201

compensating transfer 1.11 Rainy Day Savings → Everyday Checking: HTTP 201
compensating transfer 0.50 Rainy Day Savings → Everyday Checking: HTTP 201